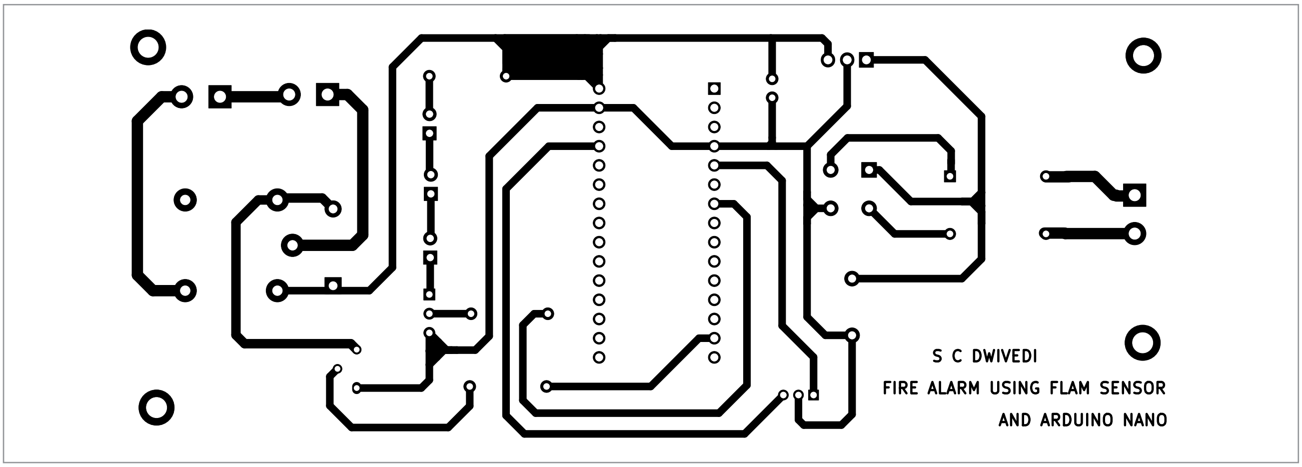

Fig. 4: Actual-size PCB layout of the circuit

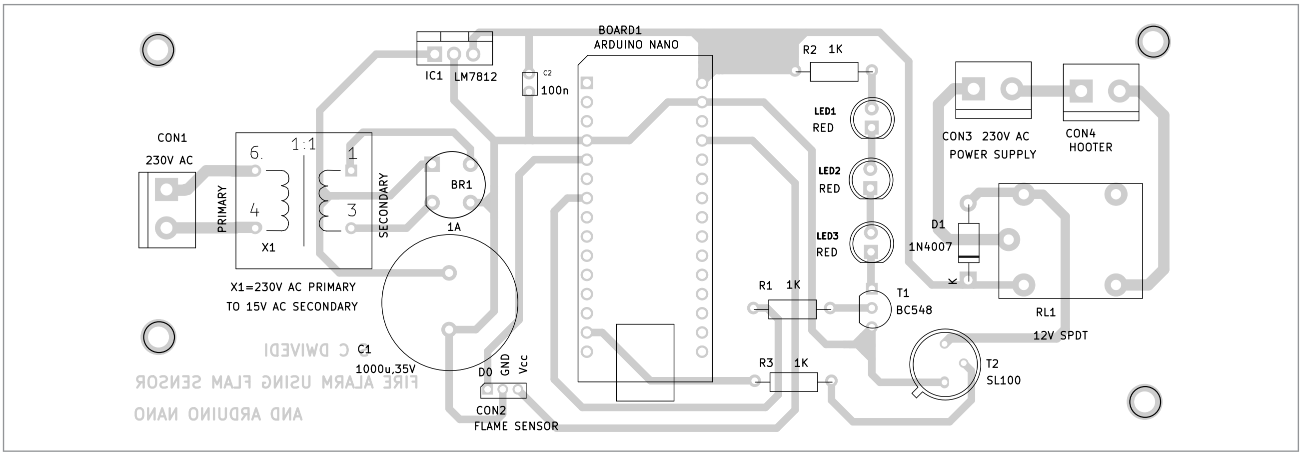

Fig. 5: Component layout of the PCB

PARTS LIST

Semiconductors:	
IC1	- LM7812, 12V voltage regulator
T1	- BC548 NPN transistor
T2	- SL100 NPN transistor
BR1	- 1A bridge rectifier
LED1-LED3	- 5mm red LED
D1	- 1N4007 rectifier diode
Resistors (all 1/4-watt, ±5% carbon):	
R1-R3	- 1-kilo-ohm
Capacitors:	
C1	- 1000µF, 35V electrolytic
C2	- 100nF ceramic disk
Miscellaneous:	
Hooter	- 230V AC operated
RL1	- 12V SPDT relay
CON1-CON3	- 2-pin connectors
Sensor1	- Flame sensor module
X1	- 230V AC primary to 15V, 500mA secondary transformer

Circuit and working

Fig. 2 presents the circuit diagram for the fire alarm using a flame sensor and Arduino Nano. The system is powered by a step-down transformer (X1), a bridge rectifier (BR1), a 12V voltage regulator (LM7812), and a flame sensor module connected via CON2. It includes a 12V SPDT relay (RL1), three LEDs (LED1 through LED3), and two transistors (BC548 and SL100), along with some other components.

The circuit operates on 12V DC, which is derived from the 15V, 500mA secondary output of transformer X1. The 230V AC mains supply connects to the primary of X1 through connector CON1. On the secondary side, the 15V AC is rectified by BR1, filtered by capacitor C1, and regulated to 12V via IC LM7812 to power the circuit.

The flame sensor module, the core of the circuit, has three pins: Vcc, GND, and output (D0). These are connected to the 5V, GND, and D2 pins of the Arduino Nano, respectively.

The flame sensor detects fire or other light sources within a wavelength range of 760nm to 1100nm, making it essential for safety applications. The sensor uses the YG1006, a high-speed, highly sensitive NPN silicon phototransistor, to detect flames.